

and no unnecessary fillers

- Include details in query to get more relevant answers
- Include examples of the desired behaviour of the model

Output indicator: This instructs the model about the desired output creation. The indicator:

- Gives format guidance to model
- Supports content
- Continues to engineer the prompt until you achieve your desired results
- Assists model in generating more helpful solutions

Pillars of prompt engineering

The following are the pillars of prompt engineering techniques.

- **Examples:** These act as a tool for fine-tuning LLM models. They help in getting the desired response.
- **Format:** This describes user queries to receive the output results. It helps in generating productive outcomes from the LLM models.
- **Direction:** This helps in narrowing responses for effective prompt output.
- **Parameters:** These help in controlling the responses generated by LLM models.
- **Chaining:** This connects multiple GenAI calls to complete the desired task.

Principles of prompt engineering

The main principles of prompt engineering to optimise responses are:

- **Clarity:** Clearly communicate what content or information is most important. A good prompt enables the LLMs to respond with more accuracy.
- **Structure the prompt:** Start by defining its role, give context/ input data, and then provide the instruction.
- **Examples:** Use specific examples to help the LLMs to narrow their focus and generate more accurate results.
- **Constraints:** Use constraints to limit the scope of the model's output. This helps in receiving the desired response and avoids unwanted or unnecessary information.
- **Tasks:** Break down complex tasks into a sequence of simpler and compact prompts. Make it simpler for LLMs to handle the data to comprehend each enquiry and generate meaningful answers.
- **Specific:** Leave as little room for interpretation as possible. Restrict the operational space.

Prompt engineering framework

There are a few prompt engineering frameworks that perform prompt engineering tasks on AI models. These are Longchain, Semantic Kernel,

Guidance, NeMo Guardrails, Fastrag, and LLamaIndex. Figure 1 depicts the prompt engineering framework.

Objectives and goals: The first step in the prompt engineering process is understanding the problem. Understanding the goal of the conversation and the objectives to be achieved are completed during this step. It addresses various problems like question-answering, text generation, sentiment analysis, etc. Analysis of tasks and constraints, and prediction of model's responses to various inputs are done.

Prompt creation: Initial prompt creation and its importance are covered during this step. Relevant data and meaningful keywords to train the model for generating the desired response/ output are identified.

Response evaluation: During this step, effective use of the initial prompt is made. Clear and concise instructions are provided in the prompt. These prompts are kept simple and clear by using plain language. Identification of discrepancies, understanding limitations, and refining of prompts are done during this step.

Iterate and refine: Assessment of the model's output and its accuracy in terms of achieving the desired objective are done during this step. Any misalignment or flaws are

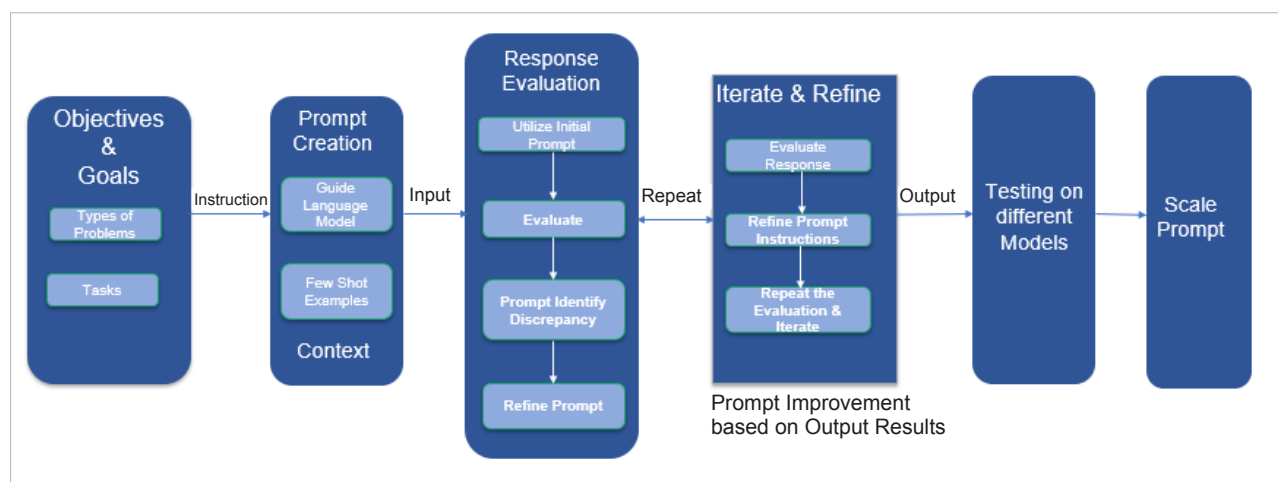


Figure 1: Prompt engineering framework